

Hands on 1

Create a Spring Web Project using Maven

The Spring Initializr web form is used to generate a new Spring project. It includes sections for Project, Language, Spring Boot version, Project Metadata, and Dependencies.

Project

- ☐ Gradle - Groovy
- ☐ Gradle - Kotlin
- ☒ Maven

Language

- ☒ Java
- ☐ Kotlin
- ☐ Groovy

Spring Boot

- ☐ 4.1.1 (SNAPSHOT)
- ☒ 4.1.0
- ☐ 4.0.8 (SNAPSHOT)
- ☐ 4.0.7
- ☐ 3.5.16

Project Metadata

Group:

Artifact:

Package name:

Packaging: ☒ Jar ☐ War

Configuration: ☒ Properties ☐ YAML

Dependencies

Spring Boot DevTools **DEVELOPER TOOLS**
Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

Spring Web **WEB**
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Buttons: GENERATE CTRL + G, EXPLORE CTRL + SPACE, ...

The IDE shows the SpringLearnApplication.java file with the following code:

```
src > main > java > com > cognizant > spring-learn > J SpringLearnApplication.java > ...
15 package com.cognizant.spring-learn;
16
17 import org.slf4j.Logger;
18 import org.slf4j.LoggerFactory;
19 import org.springframework.boot.SpringApplication;
20 import org.springframework.boot.autoconfigure.SpringBootApplication;
21
22 @SpringBootApplication
23 public class SpringLearnApplication {
24
25     private static final Logger LOGGER =
26         LoggerFactory.getLogger(SpringLearnApplication.class);
27
28     Run[Debug]
29     public static void main(String[] args) {
30
31         LOGGER.info(msg: "Application Started");
32         SpringApplication.run(primarySource: SpringLearnApplication.class, args);
33
34         LOGGER.info(msg: "Application Finished Loading");
35
36     }
37
38 }
```

The terminal output shows the following logs:

```
PS C:\Users\chava\Downloads\spring-learn\spring-learn> mvn spring-boot:run
[INFO] --- spring-boot:2.1.0:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []
20:07:09.804 [main] INFO com.cognizant.spring-learn.SpringLearnApplication -- Application Started
20:07:09.976 [restartedMain] INFO com.cognizant.spring-learn.SpringLearnApplication -- Application Started
```

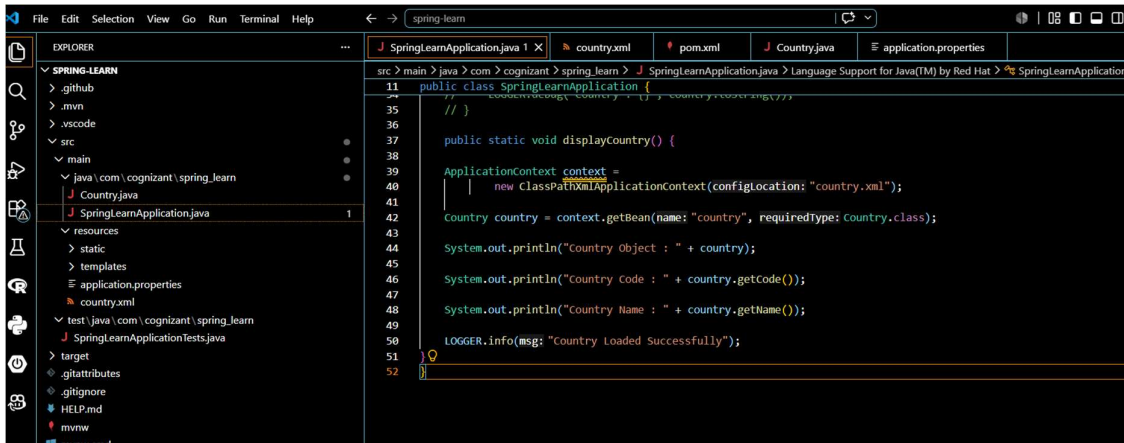
The terminal output shows the Spring Boot application running with the following logs:

```
PS C:\Users\chava\Downloads\spring-learn\spring-learn> mvn spring-boot:run
[INFO] --- spring-boot:2.1.0:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []
20:07:09.804 [main] INFO com.cognizant.spring-learn.SpringLearnApplication -- Application Started
20:07:09.976 [restartedMain] INFO com.cognizant.spring-learn.SpringLearnApplication -- Application Started
```

The application is running on port 8080. The logs show the application started successfully and is now ready to accept requests.

Hands on 4

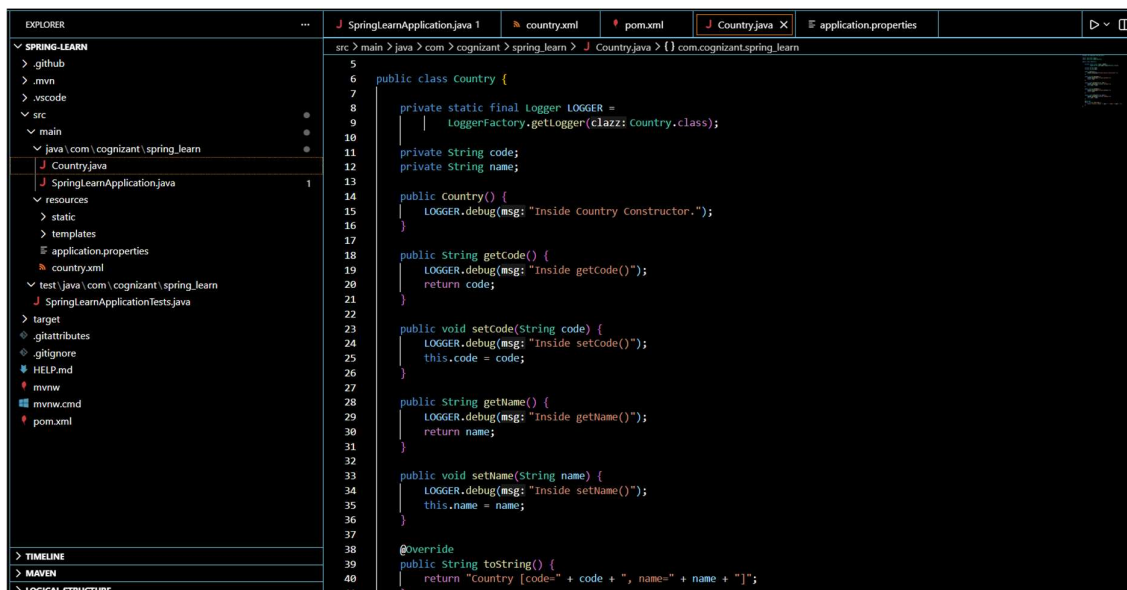
Spring Core – Load Country from Spring Configuration XML



```
src > main > java > com > cognizant > spring_learn > J SpringLearnApplication.java > Language Support for Java(TM) by Red Hat > SpringLearnApplication
11 public class SpringLearnApplication {
35 // }
36
37 public static void displayCountry() {
38
39     ApplicationContext context =
40         new ClassPathXmlApplicationContext("country.xml");
41     Country country = context.getBean(name: "country", requiredType: Country.class);
42
43     System.out.println("Country Object : " + country);
44
45     System.out.println("Country code : " + country.getCode());
46
47     System.out.println("Country Name : " + country.getName());
48
49     LOGGER.info(msg: "Country Loaded Successfully");
50 }
51
52 }
```



```
src > main > resources > country.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5
6       xsi:schemaLocation="
7         http://www.springframework.org/schema/beans
8         https://www.springframework.org/schema/beans/spring-beans.xsd">
9
10    <bean id="country"
11          class="com.cognizant.spring_learn.Country">
12
13        <property name="code" value="IN"/>
14
15        <property name="name" value="India"/>
16
17    </bean>
18
19 </beans>
```

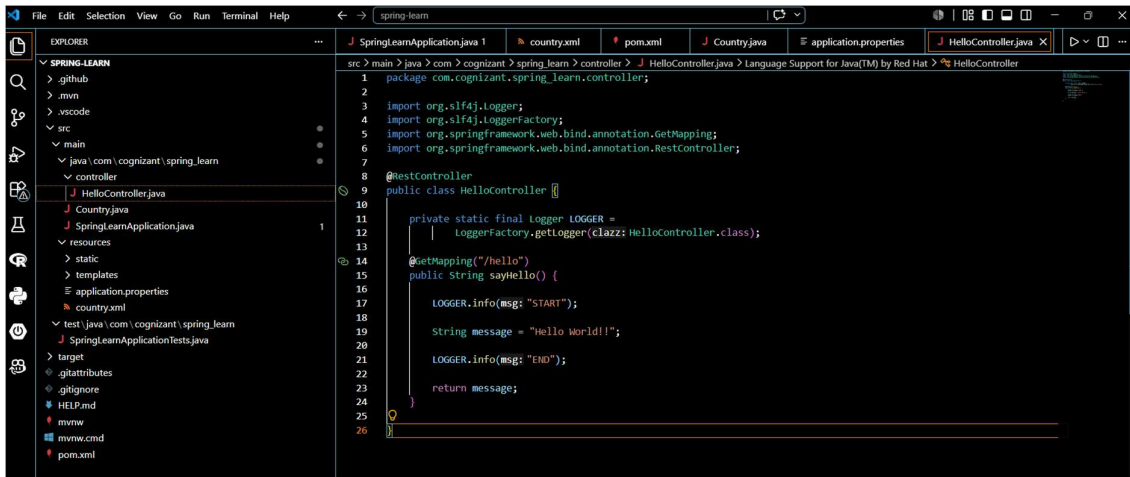
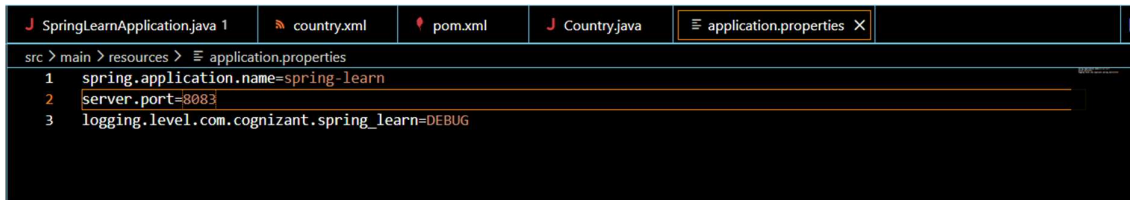


```
src > main > java > com > cognizant > spring_learn > J Country.java > () com.cognizant.spring_learn
5 public class Country {
6
7     private static final Logger LOGGER =
8         LoggerFactory.getLogger(clazz: Country.class);
9
10    private String code;
11    private String name;
12
13    public Country() {
14        LOGGER.debug(msg: "Inside Country Constructor,");
15    }
16
17    public String getCode() {
18        LOGGER.debug(msg: "Inside getCode()");
19        return code;
20    }
21
22    public void setCode(String code) {
23        LOGGER.debug(msg: "Inside setCode()");
24        this.code = code;
25    }
26
27    public String getName() {
28        LOGGER.debug(msg: "Inside getName()");
29        return name;
30    }
31
32    public void setName(String name) {
33        LOGGER.debug(msg: "Inside setName()");
34        this.name = name;
35    }
36
37    @Override
38    public String toString() {
39        return "Country [code=" + code + ", name=" + name + "]";
40    }
41 }
```

```

      \V / _ _ _ _ _ \ _ _ _ _ _ \ _ _ _ _ _ \
( ( ) \ _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ |
\ \ _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | ) ) ) )
      _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | _ _ _ _ _ | / / / /
=====|_|=====|_|/=/ / / /

```



Hello World!!

Overview GET http://localhost:8083/hc + No environment

http://localhost:8083/hello Save Share

GET http://localhost:8083/hello Send

Docs Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers (5) Test Results 200 OK 35 ms 177 B

Raw Preview Visualize

```
1 Hello World!!
```

REST - Country Web Service

File Edit Selection View Go Run Terminal Help spring-learn

EXPLORER

- SPRING-LEARN
 - .github
 - .mvn
 - .vscode
 - src
 - main
 - java.com.cognizant.spring_learn
 - controller
 - CountryController.java
 - HelloController.java

src > main > java > com > cognizant > spring_learn > controller > CountryController.java

```
1 package com.cognizant.spring_learn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.web.bind.annotation.RequestMapping;
8 import org.springframework.web.bind.annotation.RestController;
9
10 import com.cognizant.spring_learn.Country;
11
12 @RestController
13 public class CountryController {
14
15     private static final Logger LOGGER =
16         LoggerFactory.getLogger(clazz: CountryController.class);
17
18     @RequestMapping("/country")
19     public Country getCountryIndia() {
20
21         LOGGER.info(msg: "START");
22
23         ApplicationContext context =
24             new ClassPathXmlApplicationContext(configLocation: "country.xml");
25
26         Country country = context.getBean(name: "country", requiredType: Country.class);
27
28         LOGGER.info(msg: "END");
29
30         return country;
31     }
32 }
```

localhost:8083/country

Pretty-print

```
{ "code": "IN", "name": "India" }
```

Overview GET http://localhost:8083/cc + No environment

http://localhost:8083/country Save Share

GET http://localhost:8083/country Send

Docs Params Authorization Headers (8) Body Scripts Settings Cookies

Query Params

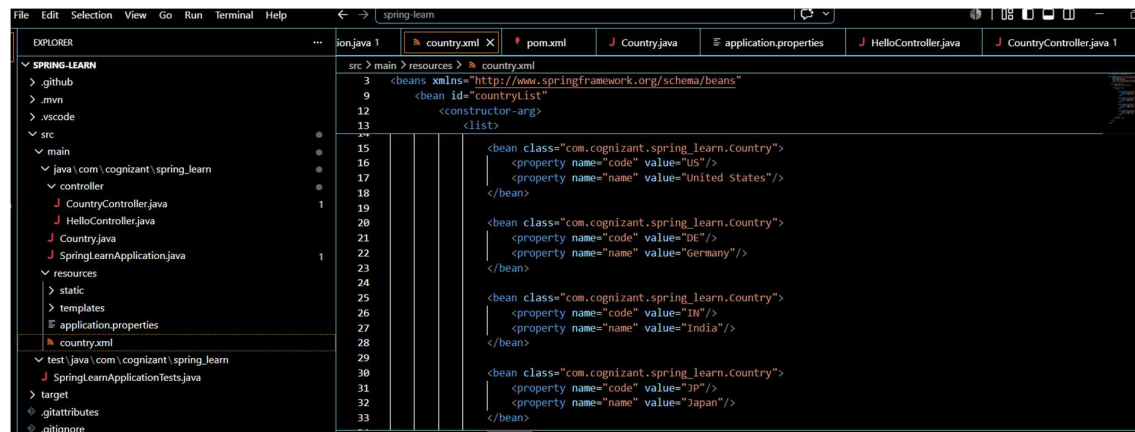
Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers (5) Test Results 200 OK 32 ms 184 B

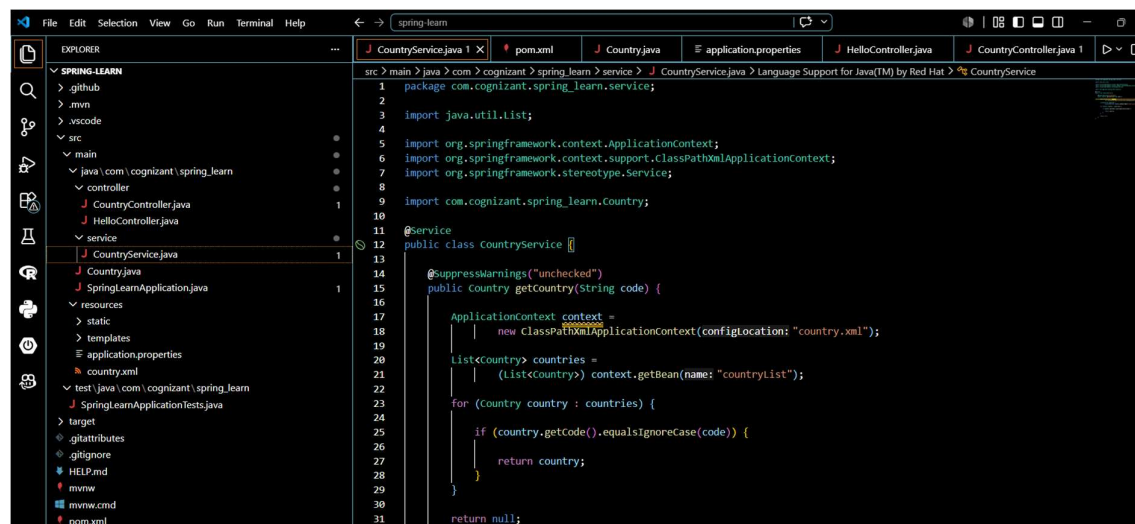
JSON Preview Visualize

```
1 { "code": "IN", "name": "India" }
```

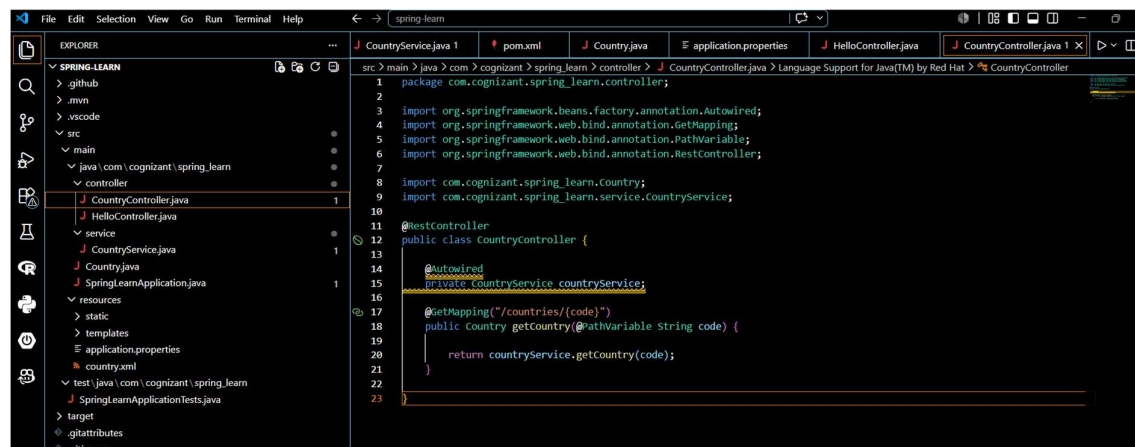
REST - Get country based on country code



```
1 <beans xmlns="http://www.springframework.org/schema/beans"
2       <bean id="countryList"
3           <constructor-arg>
4               <list>
5                   <bean class="com.cognizant.spring_learn.Country">
6                       <property name="code" value="US"/>
7                       <property name="name" value="United States"/>
8                   </bean>
9                   <bean class="com.cognizant.spring_learn.Country">
10                      <property name="code" value="DE"/>
11                      <property name="name" value="Germany"/>
12                  </bean>
13                   <bean class="com.cognizant.spring_learn.Country">
14                      <property name="code" value="IN"/>
15                      <property name="name" value="India"/>
16                  </bean>
17                   <bean class="com.cognizant.spring_learn.Country">
18                      <property name="code" value="JP"/>
19                      <property name="name" value="Japan"/>
20                  </bean>
21               </list>
22           </constructor-arg>
23       </bean>
```



```
1 package com.cognizant.spring_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.stereotype.Service;
8
9 import com.cognizant.spring_learn.Country;
10
11 @Service
12 public class CountryService {
13
14     @SuppressWarnings("unchecked")
15     public Country getCountry(String code) {
16
17         ApplicationContext context =
18             new ClassPathXmlApplicationContext("country.xml");
19
20         List<Country> countries =
21             (List<Country>) context.getBean("countryList");
22
23         for (Country country : countries) {
24
25             if (country.getCode().equalsIgnoreCase(code)) {
26
27                 return country;
28             }
29         }
30
31         return null;
32     }
33 }
```

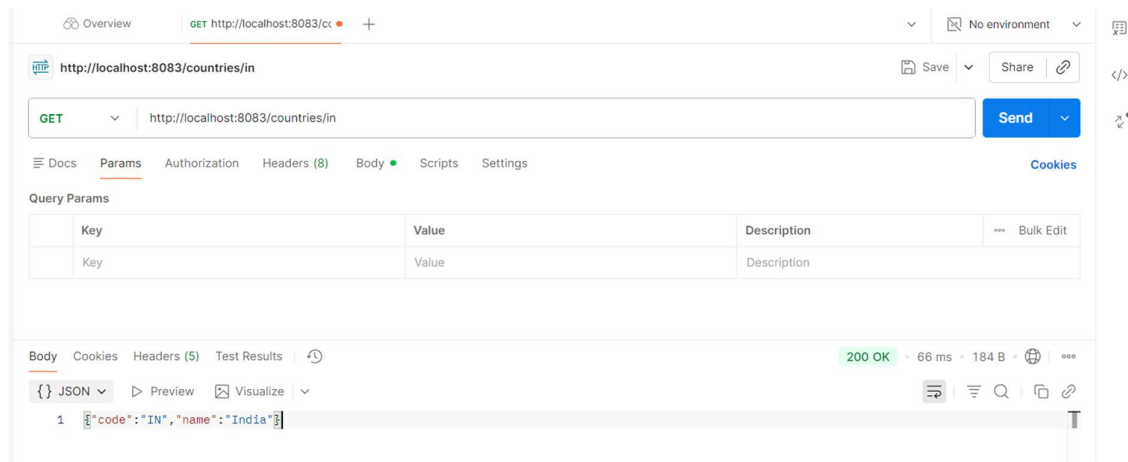


```
1 package com.cognizant.spring_learn.controller;
2
3 import org.springframework.beans.factory.annotation.Autowired;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.PathVariable;
6 import org.springframework.web.bind.annotation.RestController;
7
8 import com.cognizant.spring_learn.Country;
9 import com.cognizant.spring_learn.service.CountryService;
10
11 @RestController
12 public class CountryController {
13
14     @Autowired
15     private CountryService countryService;
16
17     @GetMapping("/countries/{code}")
18     public Country getCountry(@PathVariable String code) {
19
20         return countryService.getCountry(code);
21     }
22
23 }
```

localhost:8083/countries/DE

Pretty-print ☐

```
{"code": "DE", "name": "Germany"}
```



Auth. Service returns JWT

